

Submission Summary

Conference Name

13th International Conference on Advanced Computing - ICOAC2026

Paper ID

285

Paper Title

Predictive Decision-Support Framework for Sustainable On-Premises Data Center Resource Optimization

Abstract

On-premises data centers today operate around the clock to guarantee availability but a large share of the electricity, cooling capacity and water they have to consume goes toward servers that are barely being used. This stays hidden because the telemetry that could reveal it are typically scattered across disconnected monitoring tools that were never built to talk to each other. Our work addresses this gap through GreenOps, an observability and resource intelligence platform that unifies this data in one integrated pipeline and makes it intelligible to human operators. The system collects live telemetry (simulated in this prototype) from server, power, cooling and storage monitors and stores it with full historical context and computes real time sustainability metrics including Power Usage Effectiveness (PUE), Water Usage Effectiveness (WUE), energy cost and carbon emissions. GreenOps employs four machine learning models - CPU estimation, power estimation, workload forecasting and memory workload forecasting each trained from related telemetry. For every model, three candidate algorithms (Linear Regression, Random Forest and XGBoost) are evaluated honestly on held out data giving operators both a real-time and a near-future predictive view. Idle servers, stale storage and over-provisioned resources get flagged by a rule-based detection layer. If a server is flagged idle, a forecasting layer projects its near term and a what-if simulation engine trials possible migrations and estimates the savings involved. A priority-weighted recommendation engine then ranks these against cost, carbon and performance risk. A React based dashboard surfaces all of this, built for a human to review rather than for automation to act on unsupervised. This paper describes the system's architecture, the reasoning behind the modeling choices and results from our working prototype.

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Submission Files

Predictive Decision-Support Framework for Sustainable On-Premises Data Center Resource Optimization.pdf (910.5 Kb, 2026-10-02 14:01:28 GMT+5:30)
